“Revolutionizing CIIT Enrollment System: A Web Based Solution for Enhancing CIITzen Enrollment Experience”

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March 15, 2025

Technical Communication-GE008-Section 2

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EduTrack

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Mx. Mark Dannielle Montealegre

CIIT General Education Instructor

CIIT College of Arts and Technology

94 Kamuning Rd, Diliman

Quezon City, 1103 Metro Manila

Dear Mx. Mark Dannielle Montealegre

EduTrack aims to critically evaluate CIIT's current admission process to evaluate enhancements, covering all challenges including system crashes, lack of instructor visibility, and limited options in building the course timetable. Suggestions for enhancements including automation of requirement management, student-driven timetable building, and instructor input. Informed by the budget, process and implementation schedule, the proposal provides a complete and pragmatic course of action. The document will certainly demonstrate our solution proposition within a long-term vision for CIIT's academic processes. Please do not hesitate to provide commentary or request clarification. Any information you request will be supplemented by my main contact, Simon Andrew Labay - simon.labay@ciit.edu.ph should you like to contact him.

Thank you for your time and consideration.

Sincerely

Simon Andrew R. Labay

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Executive Summary

The “EduTrack” project proposes to restructure CIIT enrollment processes through innovative improvements to their current system weaknesses. Students who dedicatedly developed this solution designed it to overcome the system breakdowns regularly occurring and improve scheduling possibilities while revealing instructor rating scores for student enrollment decision enhancement. Three fundamental components consisting of system design along with thorough testing and stakeholder involvement will enable the implementation of MySQL-based cloud databases and optimized infrastructure alongside user-friendly interfaces to achieve stability and performance. A PHP 700,000 investment covers six months (January–June 2026) to develop this system which creates a student-friendly enrollment process that lowers operational difficulties. The proposed solution integrates feedback-based enhancement and technical security measures to establish a dependable and advanced enrollment platform for CIIT. The initiative requires support from both CIIT leadership and community stakeholders to achieve its vision of educational innovation that will guide its successful implementation.

Introduction

The present CIIT enrollment system demonstrates numerous issues through system crashes that happen often while displaying poor scheduling control and insufficient access to teaching feedback. Sixty-nine students out of 71 respondents in the survey indicated that they base their enrollment choice on teacher information. Some problems in the current system create both student dissatisfaction and operational performance challenges. The “EduTrack” project proposes a system update which teams up stability with transparency while including easy-to-use features to counter existing enrollment system challenges. The initiative streams together with CIIT’s education innovation efforts to equip students with essential enrollment decision support tools.

Problem Statement

Several important system problems affect the enrollment process at CIIT which negatively impacts user satisfaction. The system stability produces significant problems because it crashes two to three times per session prompting student frustration during enrollment. Students face additional difficulties because the system offers low customization power that prevents them from creating personalized schedules nor tracking their subjects effectively thus requiring

students to follow predefined enrollment pathways. The inability for students to view instructor evaluations alongside feedback creates a major problem in decision making because they lack the necessary information about their classes. The system performance during busy times creates slowdowns that force students to make hurried decisions for their class enrollment.

Objectives

The "EduTrack" project places specific objectives into action to handle existing issues. The system will accomplish its targets by implementing flexible scheduling that lets students construct their schedules according to their individual choices. Students can rate performance quality and assess instructor ratings in order to select their courses more intelligently. System automation of prerequisite rules will force students to follow proper course order thus preventing registration errors. The system configuration upgrade and optimized database methods will boost performance by handling bottlenecks to resolve system crashes and performance slowness. Progressive testing which involves user feedback of the system early in development ensures the product meets the needs of students along with administrative requirements.

Methodology

Progressive development methodology will guide the project implementation toward reaching established targets.

System Design & Development

A front-end development team employs existing web technologies to design an interface with user-friendly operations across different devices.

Scalability backend elements will get developed by designers in order to handle high student traffic during peak enrollment while keeping operational speed steady.

The project implements MySQL-based cloud databases for managing database operations that combines efficient data storage with secure student and instructor information security features.

Testing & Quality Assurance

Unit tests will check the functions of login as well as enrollment system components.

The execution of integration tests verifies that different system parts work together as a unit while detecting both system conflicts and data inconsistency problems.

Real users who join user acceptance testing verify that the system fulfills performance needs as well as user specifications.

Testing sessions for security will use high-traffic volumes to check system stability through performance testing for identifying data breach vulnerabilities.

Prototype Testing

Personnel will build the first prototype protocol during the early phase before sending it for evaluation to limited testers.

The system deployment process receives useful feedback that enables the team to fix missing features and enhance usability and make necessary improvements.

Hardware & Infrastructure

The system depends on local servers operating as backup data storage units that will provide dependable system performance and data protection.

Complete remote connection abilities between office staff and distant users will be created by the network configuration system.

The supplied workstations help administrators perform smooth system management tasks and troubleshooting operations.

User Training & Deployment

- The system management tools, and performance features will be trained to all administrators and stakeholders through extensive training programs.
- The system adoption and years-of-maintenance performance will benefit from user documentation which includes user manuals and technical guides.
- System enhancement will be maximized through post-deployment feedback which permits the identification of additional areas for development.

Technical Information

The user interface implements **front-end development** with responsive capabilities which enables access through desktops tablets and mobile devices by students and administrators simultaneously. Student and administrator interface navigation flows seamlessly because of the modern frameworks such as React.js or Vue.js which create an interactive user experience.

Server implementation will involve a resilient infrastructure built to manage the increased demand during busy enrollment periods with its large traffic load. The system will use Node.js or Django technologies for fast processing alongside server load-balancing to maintain operational stability.

Student records along with instructor data and course content will be stored in the structured relational database MySQL to achieve database management tasks. Database performance will improve by applying search optimizations with index enhancements included for maximum speed execution. The system will execute periodic backup schedules which protect against data loss risks.

Management Information

Timeline

The development process begins in January 2026 and extends six months until June 2026 by following different stages. The first period occurs from January to February allowing system requirement collection then moving to create database structures and design implementation. The development activities of Phase 2 lasted from March to April when developers completed front-end projects alongside back-end implementation. The evaluation of prototypes through user surveys in the third development stage serves to enhance the system. Project deployment and administrator training occur in the last phase of the initiative after testing stages are finished during June.

Stakeholder Involvement

After receiving strategic approval the CIIT Leadership maintains power to review essential decisions. Workflow testing carried out by administrators serves as the process to evaluate system needs. Users will perform prototype testing to determine system usability before offering

recommendations which will lead to system improvement. The development along with flawless deployment of technology emerges from Developers' and QA Teams' collaborative work.

Cost Information

The development of the "EduTrack" system requires an estimated PHP 700,000 budget for various programming sectors.

• Software Development (PHP 220,000):

• Front-end (PHP 100,000): UI/UX

• Back-end (PHP 120,000): Includes server setup

• Cloud Hosting & Infrastructure (PHP 150,000):

The annual expenditure for Cloud Database from the company amounts to PHP 60,000.

Domain registration services form part of this web hosting service bundle at a cost of PHP 10,000.

• Local Servers & Hardware (PHP 80,000): On-premises servers

• Testing & Training (PHP 90,000):

• Quality Assurance (PHP 50,000): Unit, integration, and security testing.

• User Training (PHP 40,000): for administrators and support staff.

• Contingency Funds (PHP 10,000): Reserved for unforeseen expenses.

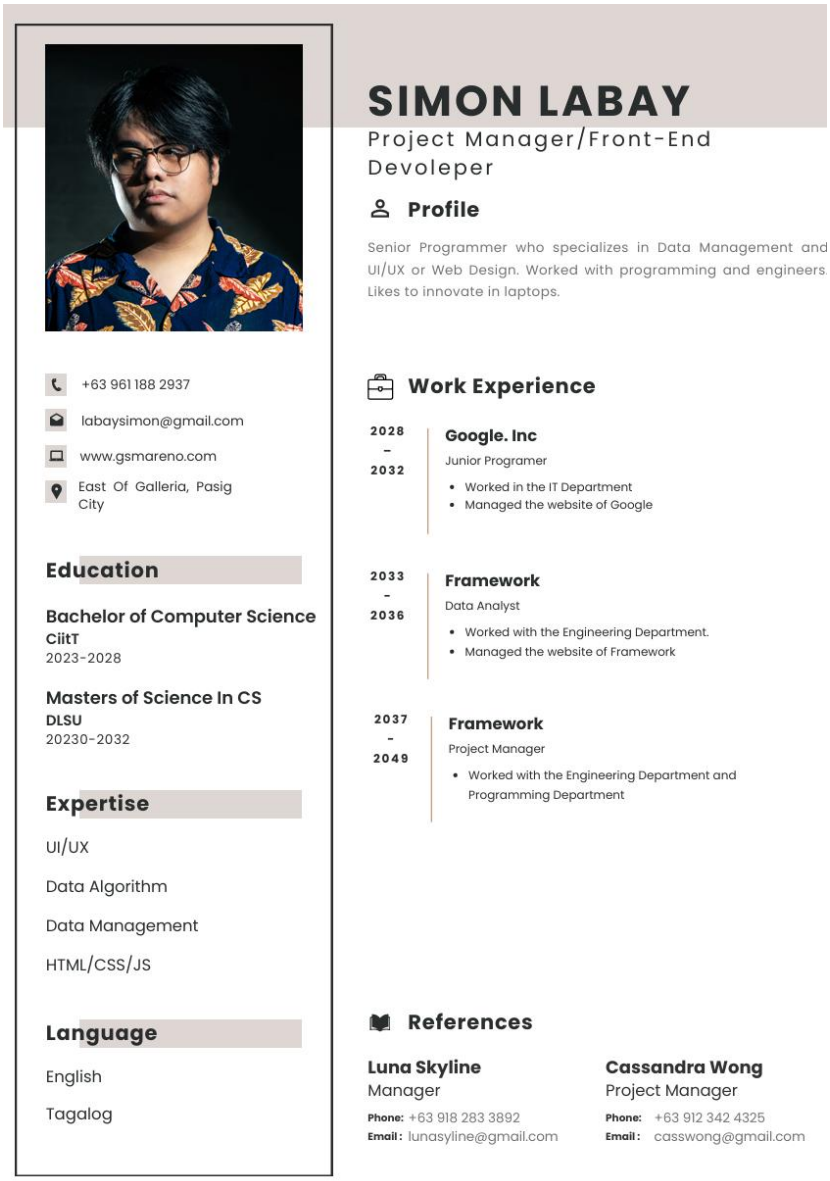
Conclusion

The proposed system improvements at CIIT's enrollment platform will create major operational benefits and better service to students and improved visibility for instructors. This project implements solutions to system failures and improves scheduling flexibility and enhances availability of instructor details to enable students with greater control resulting in decreased anxiety and improved decision quality. Through an organized development plan together with thorough testing requirements and planned budget distribution this initiative plans to transform CIIT enrollment procedures. The CIIT leadership along with its community members should join forces to turn this vision into a reality. CIIT enrollment experiences require a transformation

according to your readiness. Participate with us to create a simple enrollment system which serves students effectively.

Appendices

Resumes



SIMON LABAY

Project Manager/Front-End Devoleper

Profile

Senior Programmer who specializes in Data Management and UI/UX or Web Design. Worked with programming and engineers. Likes to innovate in laptops.

+63 961 188 2937

labaysimon@gmail.com

www.gsmareno.com

East Of Galleria, Pasig City

Education

Bachelor of Computer Science
CiitT
2023-2028

Masters of Science In CS
DLSU
20230-2032

Expertise

UI/UX

Data Algorithm

Data Management

HTML/CSS/JS

Language

English

Tagalog

Work Experience

2028 - 2032
Google. Inc
Junior Programer
- Worked in the IT Department
- Managed the website of Google

2033 - 2036
Framework
Data Analyst
- Worked with the Engineering Department.
- Managed the website of Framework

2037 - 2049
Framework
Project Manager
- Worked with the Engineering Department and Programming Department

References

Luna Skyline
Manager
Phone: +63 918 283 3892
Email : lunasyline@gmail.com

Cassandra Wong
Project Manager
Phone: +63 912 342 4325
Email : casswong@gmail.com

ARTHUR FREDERICK N. CAMBAY

Painter & Digital Artist

+63 968 381 3505

arthur.cambay@ciit.edu.ph

www.ArthurArts.com

11 ROTC Brgy. Tatalon, Q.C.

Profile

Art Director who specializes in Fine/Traditional art and digital arts. Worked with a painting studio, and Digital Arts firm, and a company.

Education

Bachelor of Multimedia Arts
CIIT
2023 - 2028

Expertise

Fine Arts

Digital Arts

Copywriting

Art Management

Language

English

Tagalog

Work Experience

2020 - 2023

Super Duper Art Gallery

Painter

- Worked in painting on canvas
- Concept arts

2023 - 2023

Writing for New Media

Ideation and Art Management

- Worked in creating brand playbook

2024 - 2024

Post-Production Techniques

Actor and Editior

- Worked on creating music video

2025 - 2025

Technical Writing

Visual and Graphics Designer

- Worked in creating skethces and ideas for frameworks

References

Alfred Sisley
Master Artist
Phone: 123-456-7890
Email : Aflred@Sisley.com

Harumi Honey
Art Director
Phone: 123-456-7890
Email : Harumi@Honey.com

Breschnan V. Fernandez

User Interface Designer

About Me

My goal in contributing in my group is to share the idea of what makes the interface to look clean and appealing.

Contact

09176167213

breschnan.fernandez@ciit.edu.ph

09176167213

Skills

- Web Design
- Branding
- Graphic Design
- BMMA -ANIMATION
- FREE LANCER

Language

- English
- Tagalog

CIIT college years

(2022)

BMMA - ANIMATION

Worked in previous personal collaboration

Started learning/working on Graphics designing.

(2023 - 2024)

BMMA - ANIMATION

Started learning 3D on hand

Worked on magazine design

Experience

(2020 -2021)

FREE LANCER

Digital artist

- Created a 2D avatar for discord voice
- Drawing profile pictures and cover photo artworks

(2020-on going)

FREE LANCER

- Started on experimenting different ratio
- Doing different kinds of animator art styles.

Marion Andrea Quinto Tiu

2nd Year Bachelor of Multimedia Arts with Specialization in Graphic Design

CIIT College of Arts and Technology

+639082561000 | mariontiu.error@gmail.com | errors-cryptic-art.carrd.co

Professional Experience

December 2023
Seller - *SPCP Arts & Design SI-A&D '22*

- Sold for the Arts and Design track's Christmas charity art market

January 2023
Creatives Member - *SPCP SPECTRUM '23*

- Joined the creatives team to advertise the senior's booth for the school fair

April 2023
Creatives Member - *SPCP G.I.F.T. Center for Visual Arts*

- Joined the creatives team to advertise the CFVA Visual Arts Week Art Market

April 2023
Seller - *SPCP G.I.F.T. Center for Visual Arts*

- Sold merchandise at the CFVA Visual Arts Week Art Market

May 2023
Artist - *PAGSIBOL Arts & Design Exhibit*

- Made art for the A&D art exhibit at Robinson's Galleria

May 2023
Photographer - *PAGSIBOL Arts & Design Exhibit*

- Documented and photographed the entire A&D art exhibit opening and the art exhibited at the event

February 2024
Seller - *CIIT Safezone Valentine's Week Art Market*

- Sold merchandise in for an in school event and for charity

May 2024 - Present
Community Advocate - *CIIT Safezone Advocacy Organization*

- Joined officially and currently help out different organization events and make sure that things run smoothly

June 2023 . April 2024 . July 2024
Part Time Employee - *BQA Events*

- Assisted at birthday parties and a wedding in making sure the events run smoothly

Awards

May 2023
Academic Achievement Award

Educational Background

Bachelor of Multimedia Arts: Graphic Design (2023 - Present)
- CIIT College of Arts and Technology
- 94 Kamuning Road, Diliman, Quezon City

SPCP Senior High School Arts & Design Track (2021 - 2023, Graduated May 2023)
- St. Paul College Pasig
- St. Paul Road, Barangay Ugong, Pasig City

Skills

- **Improvisation**
- **Bilingual:**
  - *Fluent Level:*
    - English
    - Filipino
- **Problem-Solving Skills**
- **Attention to Detail**
- **Research Proficiency**
- **Creative Work**
  - **Basic Graphic Design**
  - **Basic Photoshop**
  - **Basic Watercolor Painting**
  - **Basic Acrylic Painting**
  - **Digital Illustration**
  - **Traditional Illustration**

Survey Questionnaires

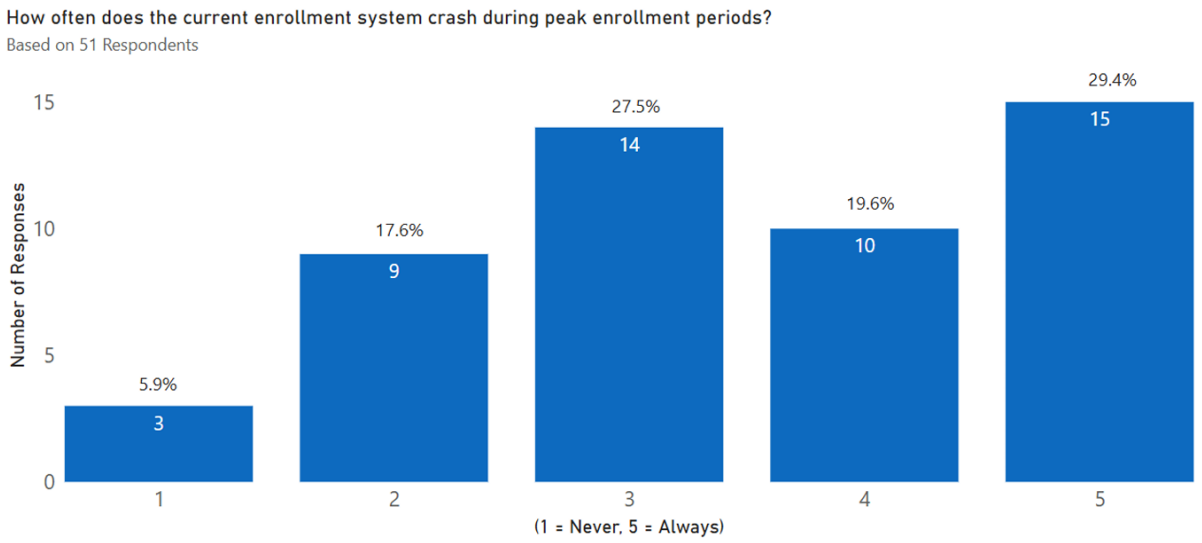
1. How often does the current enrollment system crash during peak enrollment periods?
2. How satisfied are you with the current system's loading speed during enrollment?
3. How many times, on average do you experience system crashes per enrollment session? (Enter a number)
4. Is it important for you to see instructor information before enrolling?

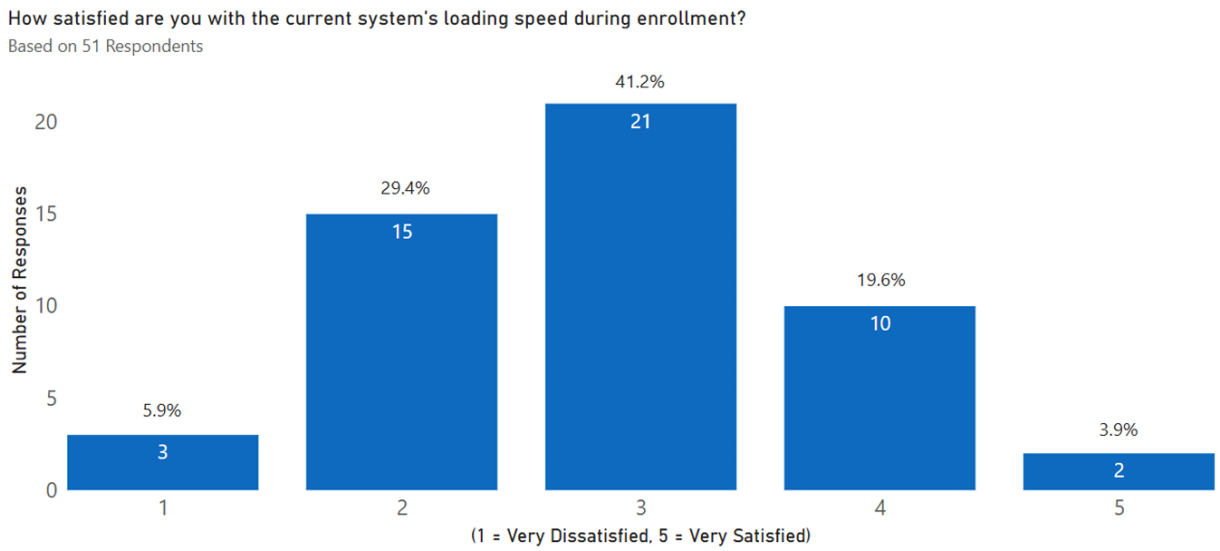
5. What specific information do you want to see? (choose 1 or more)

   Check all that apply.

   - Instructor name and subject
   - Performance and ratings
   - Student feedback quotes
   - None
   - Other
6. How likely are you to choose a subject based on instructor performance metrics?
7. How easy is it to navigate the current enrollment system?
8. What features would improve the system? (choose 1 or more)
   - Schedule customization
   - Mobile application
   - Integrated messaging system
   - Subject tracker
   - Subject prerequisite tree
9. How important is it to include student feedback ratings for instructors?
10. How many years of instructor performance data should be displayed?

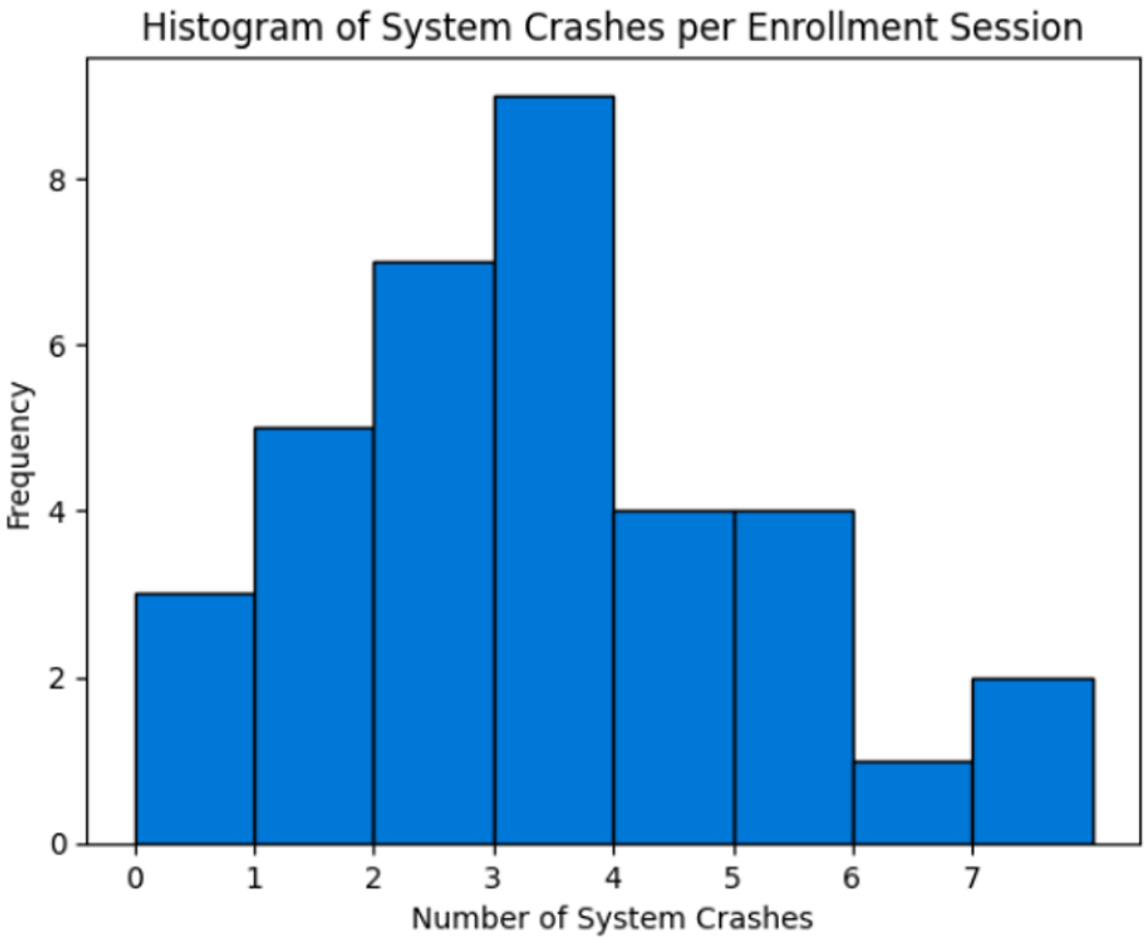
Survey Results





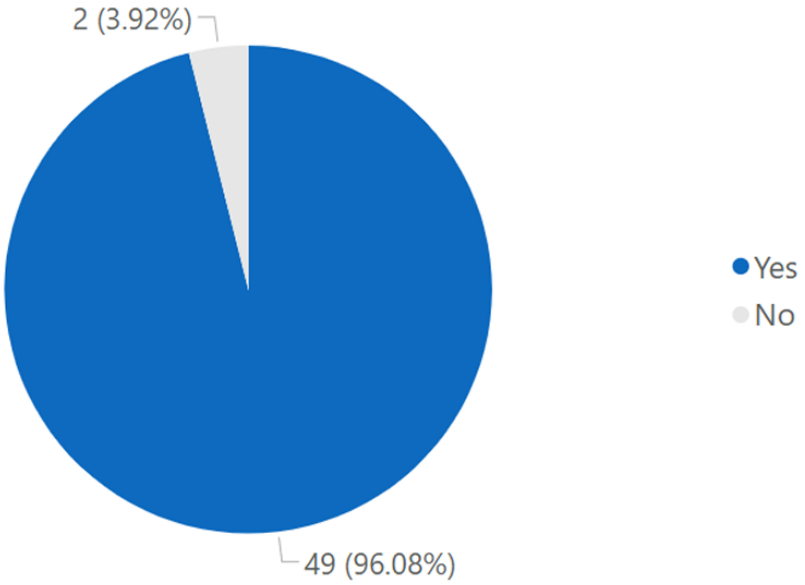
How many times, on average do you experience system crashes per enrollment session?

Based on 51 Respondents



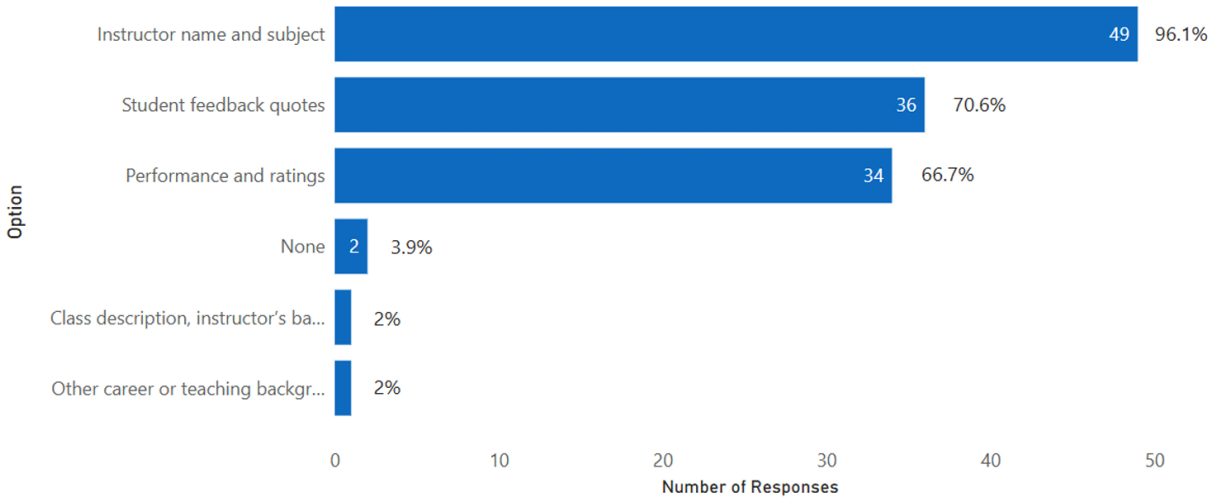
Is it important for you to see instructor information before enrolling?

Based on 51 Respondents



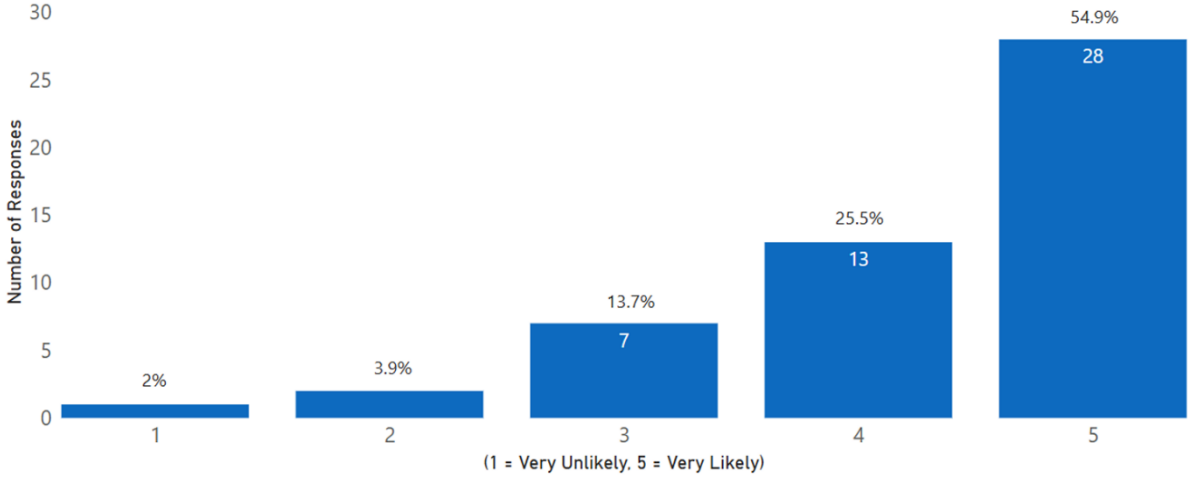
What specific information do you want to see?

Based on 51 Respondents



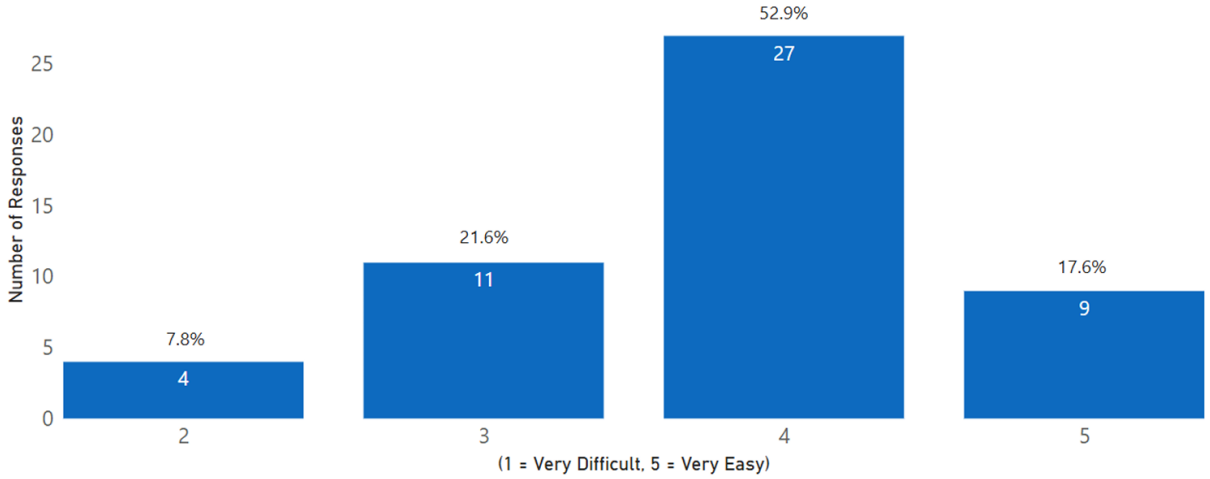
How likely are you to choose a subject based on instructor performance metrics?

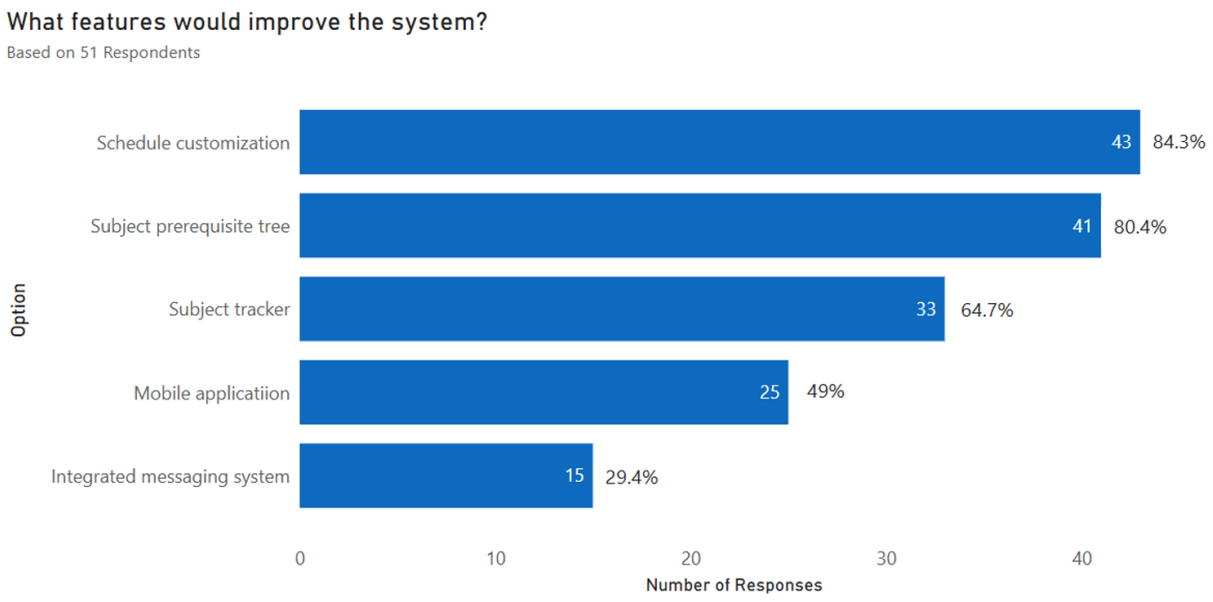
Based on 51 Respondents



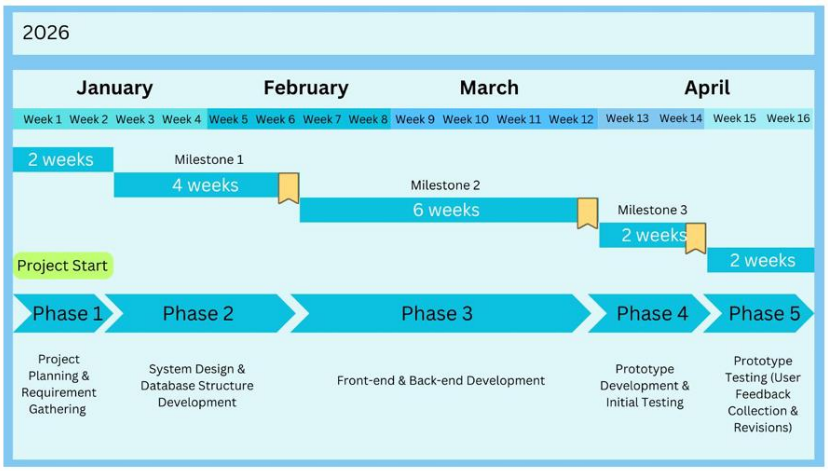
How easy is it to navigate the current enrollment system?

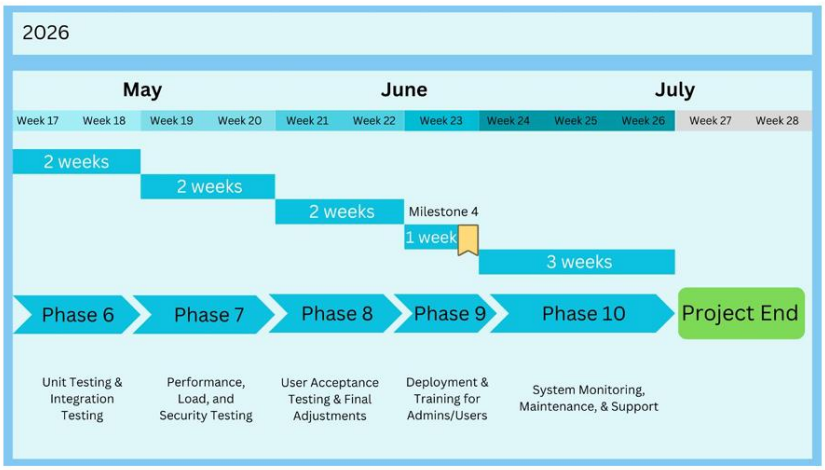
Based on 51 Respondents





Schedule and Timeline Breakdown

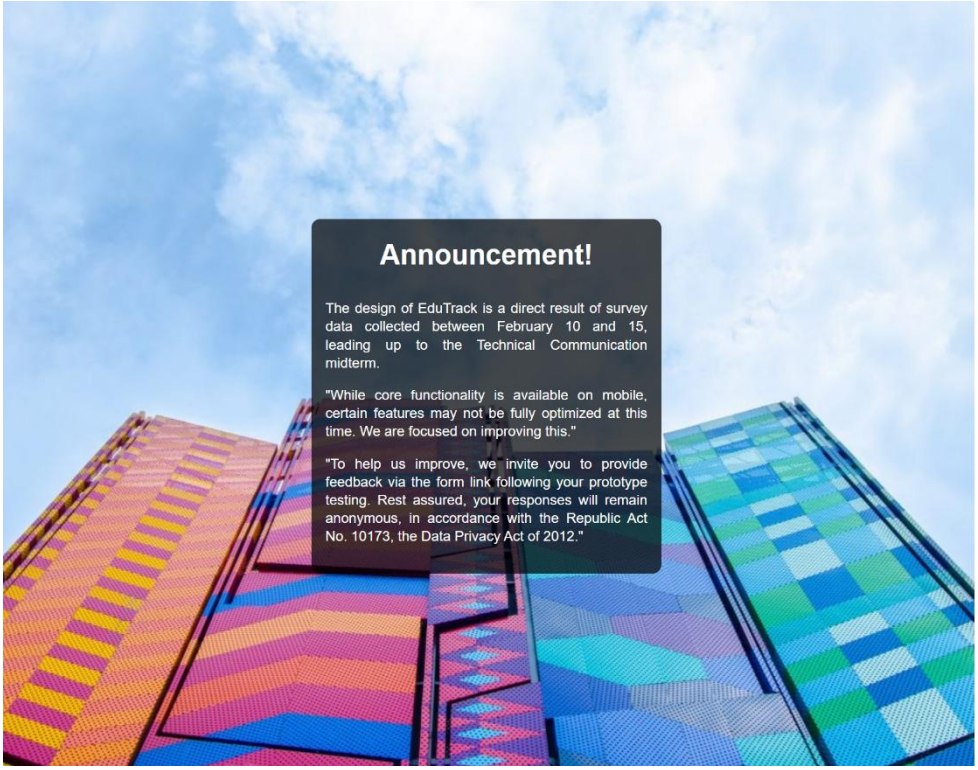


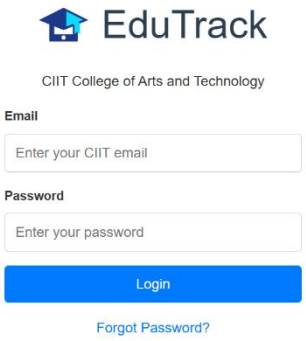


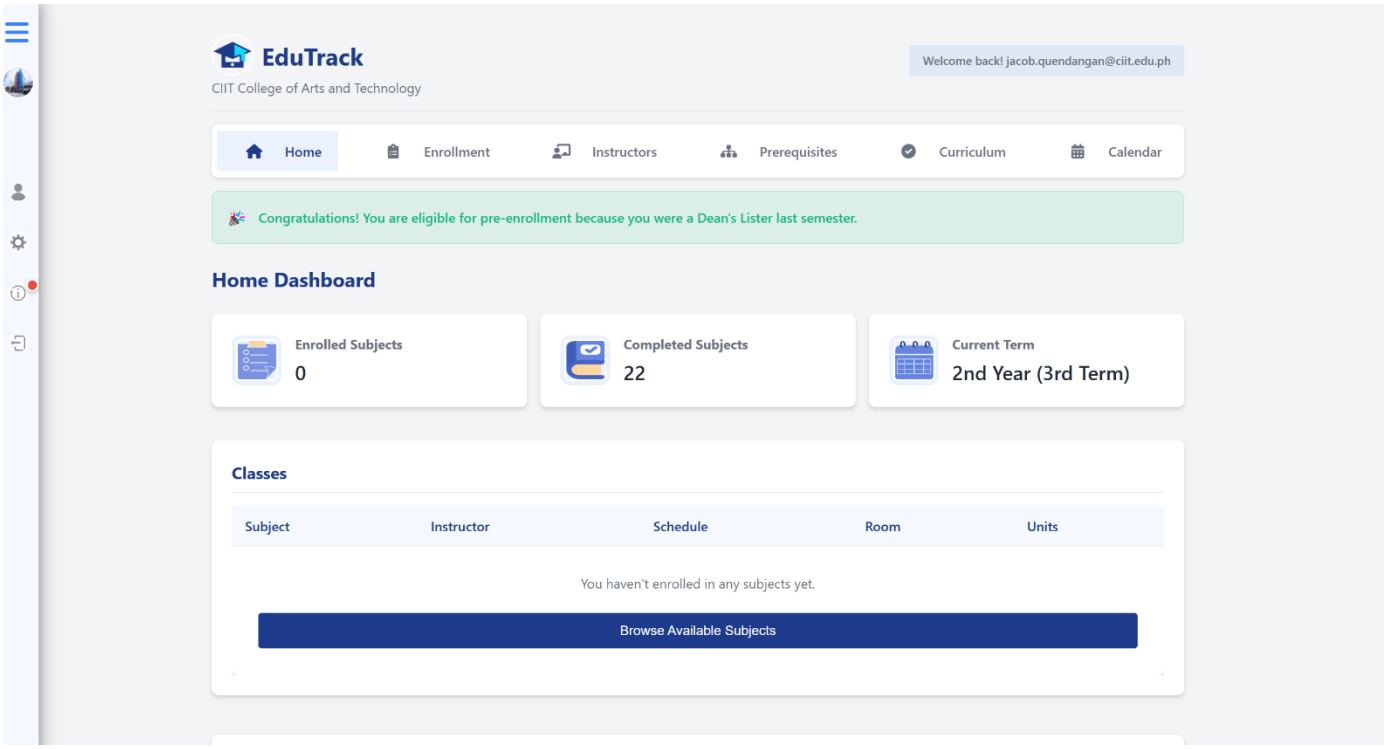
Costs Breakdown

Component/Activity		Estimated Cost (PHP)	Validation
Software Development	Front-end	100,000.00	Development of web-based enrollment system with database integration
	Back-end	120,000.00	
Cloud Database Hosting (1 year)		60,000.00	Based on AWS RDS (MySQL) or Google Cloud SQL (₱5,000/month for moderate usage)
Domain & Web Hosting (1 year)		10,000.00	Includes domain registration and VPS hosting
Database Design & Implementation		80,000.00	Structured relational database with optimized queries
Hardware	Local Server	80,000.00	For self-hosted database and system backup
	Workstations	50,000.00	2–3 desktop PCs for admin staff
	Network Setup	20,000.00	Routers, switches, and configuration
Testing & Quality Assurance	Unit Testing	15,000.00	Ensures that individual components (e.g., login, enrollment, payment) work correctly.
	Integration Testing	15,000.00	Confirms that different system modules work together properly.
	User Acceptance Testing	10,000.00	Verify that the system meets user needs and is easy to use.
	Performance & Load Testing	10,000.00	Ensures the system can handle multiple users and heavy data loads.
	Security Testing	10,000.00	Identifies vulnerabilities to prevent hacking or data breaches.
Deployment & Training		40,000.00	System setup, administrator and user training
Maintenance & Support (1st Year)		50,000.00	Bug fixes, updates, user support
Miscellaneous	Reports & Documentation	10,000.00	Ensure all stakeholders have clear documentation on system use, database structure, and technical aspects.
	Meetings & Coordination	10,000.00	Covers logistical costs for meetings with stakeholders (school administrators, developers, testers).
	Contingencies & Emergencies	10,000.00	A buffer fund for unexpected issues during development and deployment.
Total Estimated Budget		**700,000.00**	

Sample Mockups/ Prototype/ Design Layout






EduTrack
CIIT College of Arts and Technology
Welcome back! jacob.quendangan@ciit.edu.ph
Home
Enrollment
Instructors
Prerequisites
Curriculum
Calendar
Congratulations! You are eligible for pre-enrollment because you were a Dean's Lister last semester.
Home Dashboard
Enrolled Subjects
0
Completed Subjects
22
Current Term
2nd Year (3rd Term)
Classes
Subject
Instructor
Schedule
Room
Units
You haven't enrolled in any subjects yet.
Browse Available Subjects